

PL Functions and Reeb Graphs

Lecture 16 - CMSE 890

Prof. Elizabeth Munch

Michigan State University

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Dept of Computational Mathematics, Science & Engineering

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Goals for today:

- 3.1 Filtrations on spaces, PL Functions
- 3.5 Persistence for PL Functions

Section 1

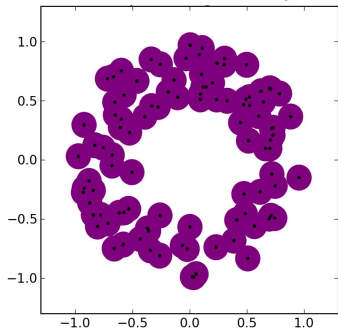
Persistence on functions

A different viewpoint for point cloud persistence

Given point cloud $P \subset \mathbb{R}^d$.

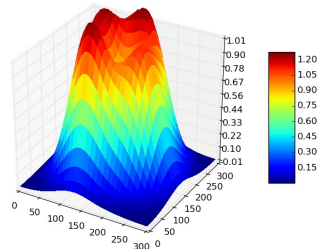
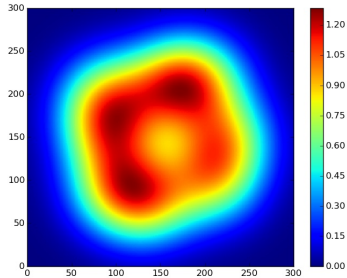
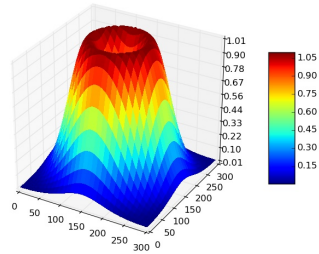
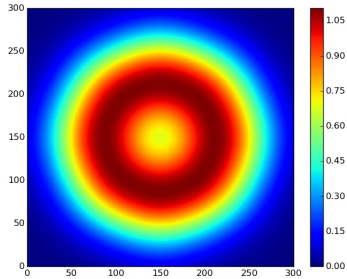
$$f : \mathbb{R}^d \rightarrow \mathbb{R}$$

$$x \mapsto \min_{y \in P} \|x - y\|$$

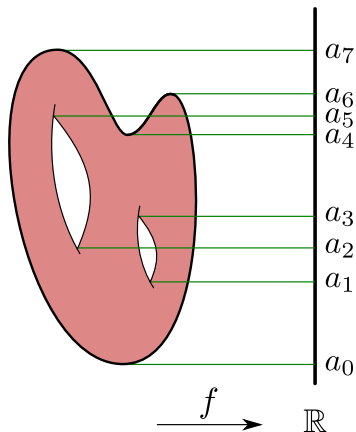


$$f^{-1}(-\infty, r] = \bigcup_{y \in P} B_r(y)$$

Different functions



Another example: “Height functions”



Persistence for functions

Let $\mathbb{X}$ be a topological space with function $f : \mathbb{X} \rightarrow \mathbb{R}$.

Let $\mathbb{X}_a = f^{-1}(-\infty, a]$.

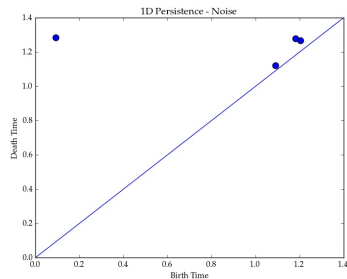
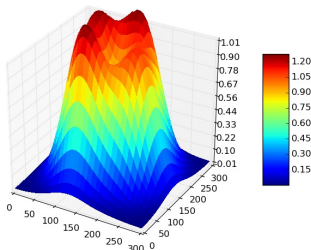
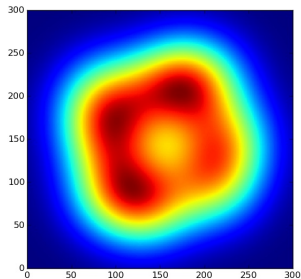
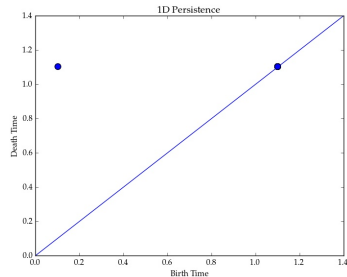
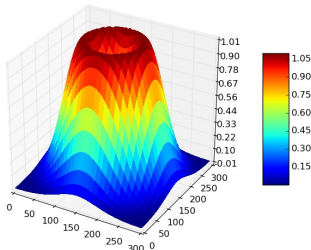
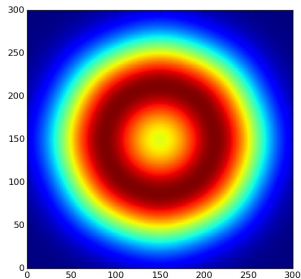
Then for $a_0 \leq a_1 \leq a_2 \leq \cdots \leq a_k$

$$\mathbb{X}_{a_0} \subseteq \mathbb{X}_{a_1} \subseteq \cdots \subseteq \mathbb{X}_{a_k}$$

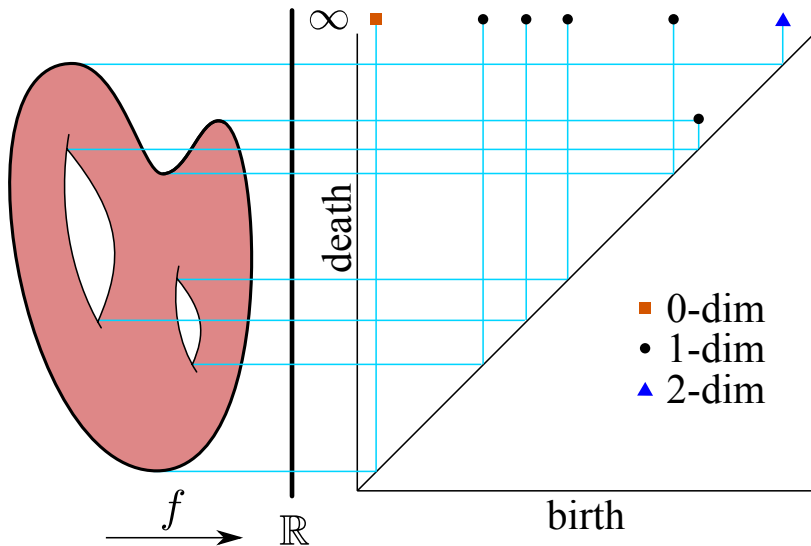
gives

$$H_p(\mathbb{X}_{a_0}) \rightarrow H_p(\mathbb{X}_{a_1}) \rightarrow \cdots \rightarrow H_p(\mathbb{X}_{a_k}).$$

Example with matrix function



Example with torus height function



Section 2

Piecewise Linear functions

PL Functions

Given simplicial complex K . A function

$$f : V(K) \rightarrow \mathbb{R}$$

gives a piecewise linear (PL) function

$$f : |K| \rightarrow \mathbb{R}$$

$$x \mapsto \sum_{v \in V(K)} (b_i(x) f(v)).$$

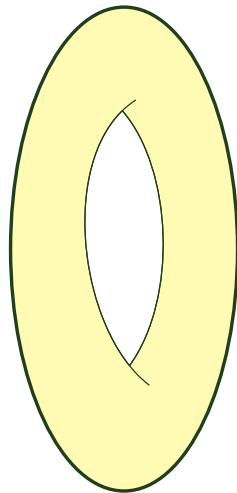
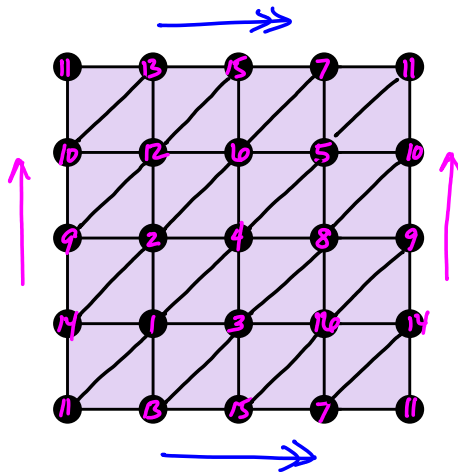
The function is *generic* if

$$f(u) \neq f(v) \quad \forall u \neq v \in V(K).$$

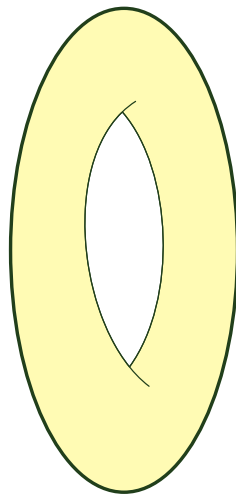
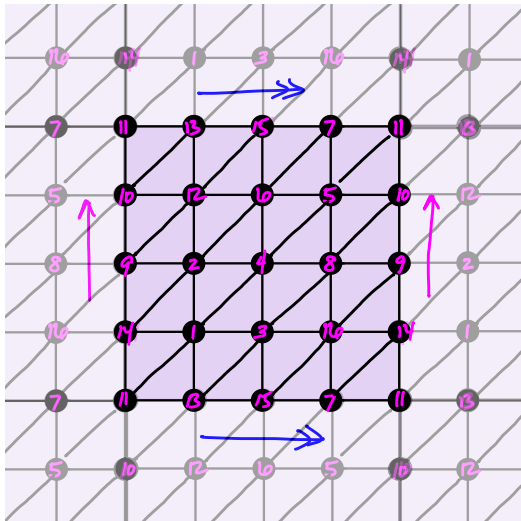
Sort the vertices so that

$$f(u_1) < f(u_2) < \cdots < f(u_n).$$

Square representation of a torus



Square representation of a torus



Level set

Given (PL) $f : |K| \rightarrow \mathbb{R}$.

A sublevel set:

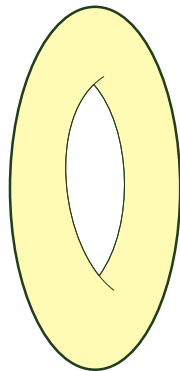
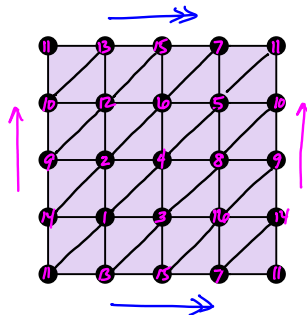
$$f^{-1}(-\infty, a]$$

A superlevel set:

$$f^{-1}[a, \infty)$$

A level set:

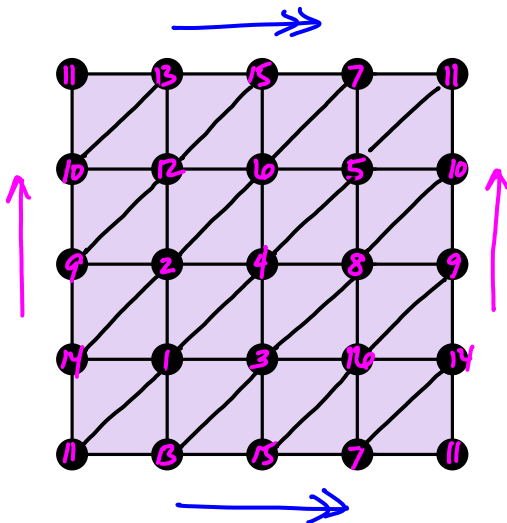
$$f^{-1}(a)$$



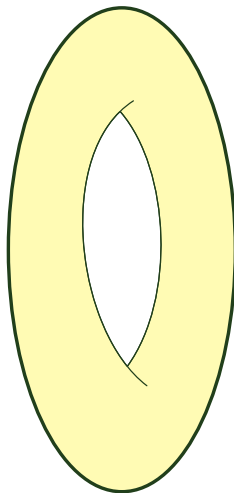
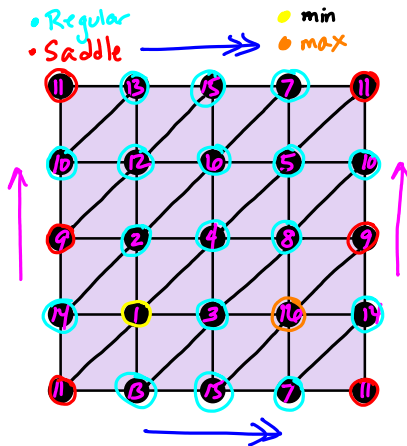
Try it:

Draw the following levelsets:

- $f^{-1}(4.5)$
- $f^{-1}(10.5)$
- $f^{-1}(9)$



Matching this example with the height function



Section 3

Reeb graphs

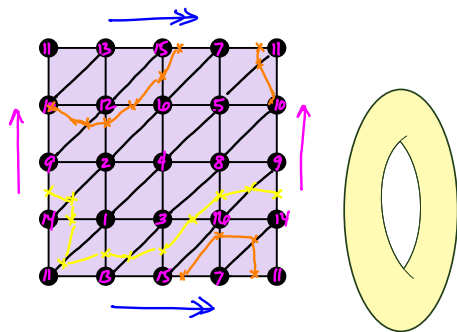
Recall: Connected Components

A space X is disconnected if there are disjoint open sets U and V with $X = U \cup V$. A space is connected otherwise.

A connected component is a maximal subset that is connected.

Contours vs level sets

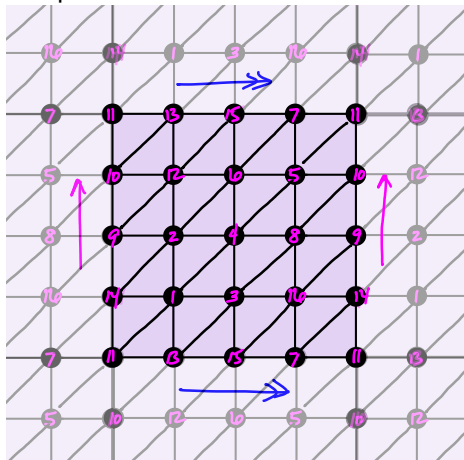
A contour is a connected component of a level set.



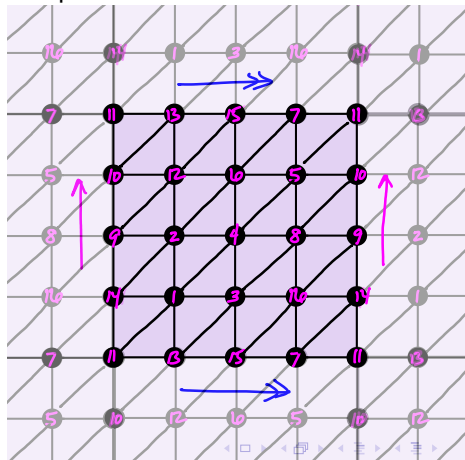
How do the number of contours change as the level set function value changes?

Try it

Sketch the level set $f^{-1}(8)$. What shape is it? How many connected components does it have?



Sketch the level set $f^{-1}(9)$. What shape is it? How many connected components does it have?



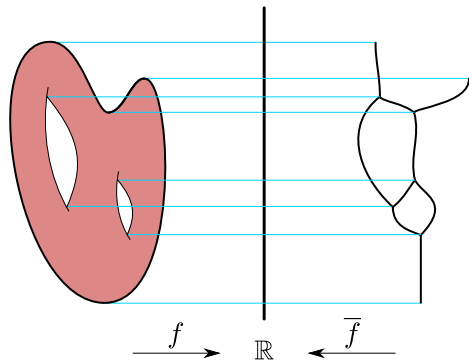
Equivalence relation

A binary relation $\sim$ on a set $\mathbb{T}$ is an equivalence relation if for every $a, b, c \in \mathbb{T}$:

- $a \sim a$ (reflexivity)
- $a \sim b$ iff $b \sim a$ (symmetry)
- If $a \sim b$ and $b \sim c$ then $a \sim c$

The equivalence class of a under $\sim$ is $[a] = \{x \in \mathbb{T} \mid x \sim a\}$.

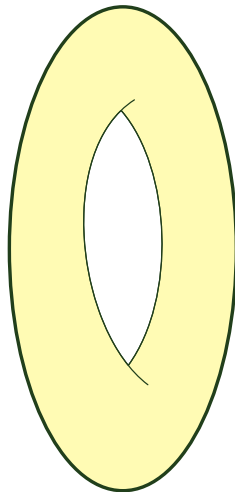
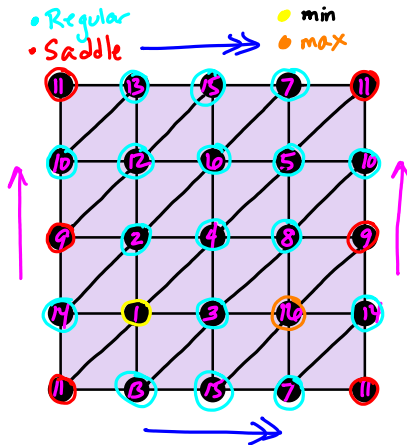
Reeb graph equivalence relation



Given a function $f : X \rightarrow \mathbb{R}$.
Define an equivalence relation $\sim$ by
 $s \sim y$ iff

- $f(x) = f(y) = \alpha$
- x and y are in the same connected component of the level set $f^{-1}(\alpha)$.

Equivalence relation example



Quotient space

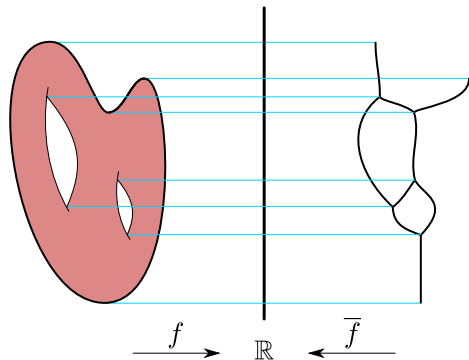
Given a topological space $(\mathbb{T}, T)$ and an equivalence relation $\sim$ defined on the set $\mathbb{T}$, the quotient space $(\mathbb{S}, S)$ consists of

- points $\mathbb{S} = \mathbb{T} / \sim = \{[a] \mid a \in \mathbb{T}\}$
- open sets $S = \{U \subseteq \mathbb{S} \mid \{a \mid [a] \in U\} \in T\}$

V.2:

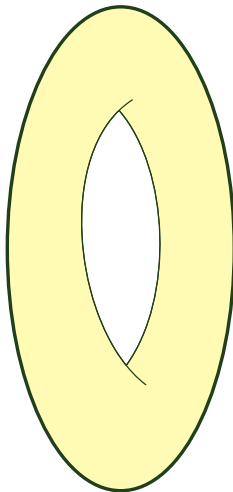
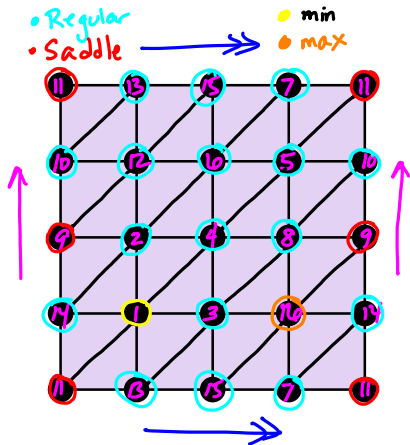
- Get a map $q : \mathbb{T} \rightarrow \mathbb{S}, a \mapsto [a]$
- $U \subseteq \mathbb{S}$ open iff $q^{-1}(U)$ is open

Reeb graph



- Let $[x]$ denote the equivalence class of $x \in X$.
- The Reeb graph R_f of $f: X \rightarrow \mathbb{R}$ is the quotient space $X / \sim$.
- Let $\Phi: X \rightarrow R_f$, $x \rightarrow [x]$ be the quotient map.

Computing the Reeb graph



Nice enough assumptions

If the input is nice enough, then R_f is a graph.....

- Morse function on a compact manifold.
- PL function on a compact polyhedron
- Others.....

Their assumptions

Require that $f : X \rightarrow \mathbb{R}$ is levelset tame:

- each level set $f^{-1}(a)$ has a finite number of components
- each component is path connected
- f is of Morse type (*Defn 4.14: each levelset has finitely generated homology groups, finitely many critical values, each interval between critical values is regular*)

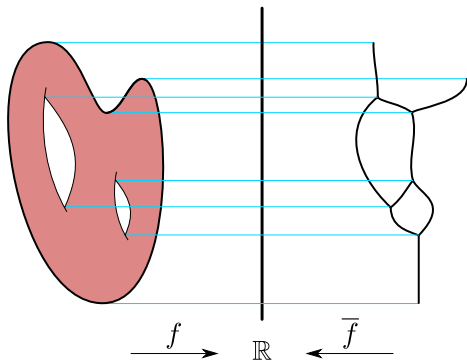
Reeb graph as a topological summary

With a bit of a lie



Figure: Pascucci et al

Induced map $\tilde{f} : R_f \rightarrow \mathbb{R}$



Lazy notation: Just write $f : R_f \rightarrow \mathbb{R}$

Another example

Compute the Reeb graph of the following example:

